

Funnels and Parameterization

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Multilevel Models

For group-level parameters $\theta_1, \dots, \theta_J$, a hierarchical prior shares information through a common location and scale.

$$\theta_j \mid \mu, \sigma_\theta \sim \text{normal}(\mu, \sigma_\theta), \quad j = 1, \dots, J$$

$$\sigma_\theta \sim p(\sigma_\theta)$$

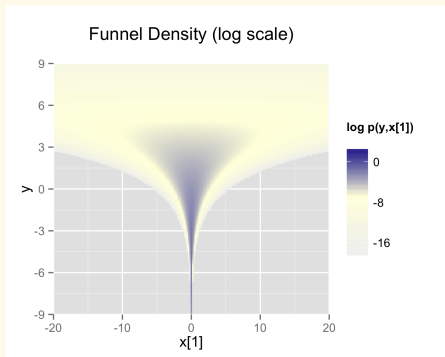
$$\theta_j = \mu + \sigma_\theta z_j, \quad z_j \sim \text{normal}(0, 1)$$

- Neal's funnel isolates the geometry created when one scale controls many effects.
- Small σ_θ pulls group effects toward μ ; large σ_θ lets them differ.
- Parameterization choice: sample θ_j directly, or sample standardized z_j and transform.

Multilevel Models and Neal's Funnel

- Neal's Funnel: extreme example of a challenging hierarchical prior

$$y \sim \text{normal}(0, 3), \quad x_n \sim \text{normal}(0, \exp(y/2)), \quad n = 1, \dots, 9.$$



- Canonical example has 9 lower-level parameters $x_1, \dots, x_9$.
- One population-scale parameter y controls their prior scale.
- To explore neck of the funnel: small steps on x -axis, large steps on y -axis.
- To explore mouth: large steps on x -axis, small steps on y -axis.

Funnel Example: Stan Implementations

Centered parameterization

```
parameters {  
  real y;  
  vector[9] x;  
}  
model {  
  y ~ normal(0, 3);  
  x ~ normal(0, exp(y / 2));  
}
```

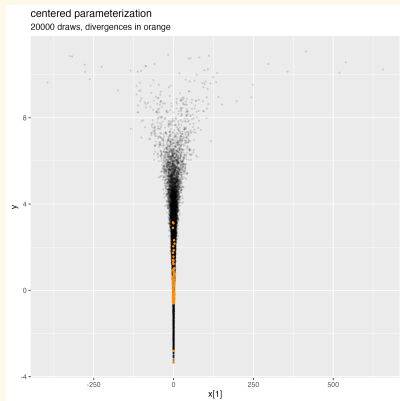
Non-centered parameterization

```
parameters {  
  real y_std;  
  vector[9] x_std;  
}  
transformed parameters {  
  real y = 3 * y_std;  
  vector[9] x = exp(y / 2) * x_std;  
}  
model {  
  y_std ~ std_normal();  
  x_std ~ std_normal();  
}
```

Low-data Regime

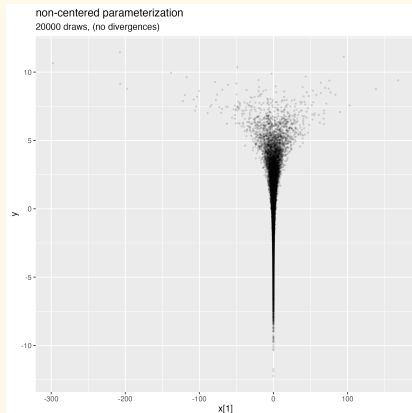
Centered

- Cannot explore neck of funnel.
- Many divergences.
- Low-data regime is prior-dominated.



Non-centered

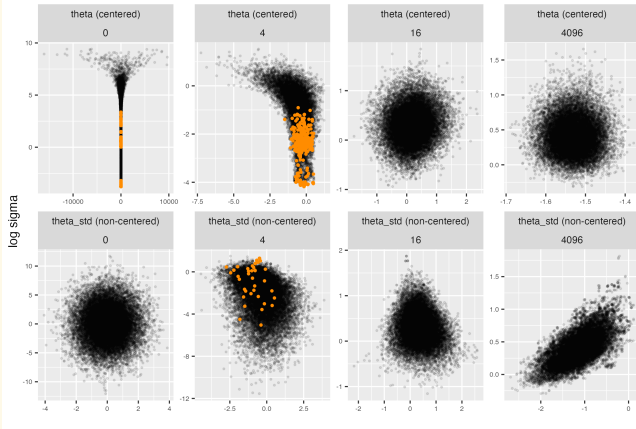
- Explores further.
- No divergences in this example.
- Works well when group effects are weakly identified.



What Happens as Data Grow?

centered vs. non-centered parameterization

hierarchical logistic regression with 10 groups, intercept only, 20000 draws, divergences in orange



- 10 groups, intercept-only hierarchical logistic regression.
- Fit model with 0, 4, 16, 4096 observations.
- 20,000 draws; divergences in orange.
- Low data: non-centered avoids the funnel.
- More data: non-centered starts to warp.

Varying Slope / Varying Intercept Model

Let x_n include a column of ones for the group-level intercept and additional columns for group-level predictors.

$$y_n \sim \text{normal}(\eta_n, \sigma)$$

$$\eta_n = \alpha + z_n \gamma + x_n^\top \beta_{g[n]}$$

$$\beta_j \sim \text{multi_normal}(\mu_\beta, \Sigma_\beta)$$

- $j = 1, \dots, J$ indexes groups.
- β_j is a length- K vector: intercept, slope, and other group-level coefficients.
- $\Sigma_\beta = \text{diag}(\tau) \Omega \text{diag}(\tau)$.

Varying Effects: Choose One Parameterization

Both versions define the same model-scale object used by the likelihood: `array[J]`
`vector[K]` `beta`.

Centered parameterization

```
parameters {  
  vector[K] mu_beta;  
  vector<lower=0>[K] tau;  
  cholesky_factor_corr[K] L_Omega;  
  array[J] vector[K] beta;  
}  
model {  
  matrix[K, K] L_Sigma =  
    diag_pre_multiply(tau, L_Omega);  
  
  mu_beta ~ normal(0, 1);  
  tau ~ exponential(1);  
  L_Omega ~ lkj_corr_cholesky(2);  
  
  for (j in 1:J)
```

Non-centered parameterization

```
parameters {  
  vector[K] mu_beta;  
  vector<lower=0>[K] tau;  
  cholesky_factor_corr[K] L_Omega;  
  matrix[K, J] beta_std;  
}  
transformed parameters {  
  array[J] vector[K] beta;  
  {  
    matrix[K, J] beta_mat =  
      rep_matrix(mu_beta, J)  
      + diag_pre_multiply(tau, L_Omega) * beta_std;  
    for (j in 1:J) beta[j] = beta_mat[, j];  
  }  
}  
model {  
  for (j in 1:J)
```


Common Likelihood After Either Choice

```
data {  
  int<lower=1> N;   int<lower=1> J;   int<lower=1> K;   int<lower=0> P;  
  array[N] int<lower=1, upper=J> group;  
  matrix[N, P] z_pop;           // population-level predictors  
  matrix[N, K] x_grp;           // first column is 1 for varying intercept  
  vector[N] y;  
}  
  
parameters {  
  real alpha;   vector[P] gamma;   real<lower=0> sigma;  
  // plus one varying-effects block: centered or non-centered  
}  
  
model {  
  alpha ~ normal(0, 1);   gamma ~ normal(0, 1);   sigma ~ exponential(1);  
  {  
    vector[N] eta = rep_vector(alpha, N) + z_pop * gamma;  
    for (n in 1:N) eta[n] += x_grp[n] * beta[group[n]];  
    y ~ normal(eta, sigma);  
  }  
  // plus priors from the selected varying-effects block  
}
```

Same Model, Different Geometry

- Centered:
 - sample group coefficients directly from the population distribution.
 - often better when each group has many observations.
- Non-centered:
 - sample standardized group coefficients.
 - transform by population location, scale, and correlation.
 - often better in weak-data or prior-dominated regimes.

Choice depends on amount of information per group-level predictor plus what you know about the data.

- [A General Framework for the Parametrization of Hierarchical Models, Papaspiliopoulos et al.](#)
- [Hamiltonian Monte Carlo for Hierarchical Models, Betancourt and Girolami](#)